MySQL Practice Scripts

1. Overview

This repository contains a structured collection of MySQL SQL scripts designed for learning and practising core database concepts. Each file is organized by topic and progressively covers fundamental to advanced MySQL features.

The scripts are suitable for beginners learning SQL as well as intermediate users who want to reinforce their understanding of relational database concepts.

2. File Structure

The following SQL files are included in this project:

File Name	Description
`MYSQL_Clauses_&_Operators.sql`	WHERE, LIKE, IN, BETWEEN, AND/OR/NOT, IS NULL, DISTINCT, Aliases
`MYSQL_Clauses_&_Operators-ORDER_BY...sql`	ORDER BY, LIMIT, Aggregate Functions, GROUP BY, HAVING, Window Functions
`MYSQL_DDL_Commands-Hospital_Database.sql`	CREATE, ALTER, RENAME, TRUNCATE, DROP — Hospital Database
`MYSQL_DML_and_Constraints-Online_Bookstore.sql`	INSERT, UPDATE, DELETE, Constraints (PK, FK, UNIQUE, CHECK, NOT NULL)
`MYSQL_Joins_&_Buildin_Functions.sql`	INNER/LEFT/RIGHT JOIN, String, Math, and Date Built-in Functions
`MYSQL_SUBQUERIES.sql`	Single Row, Multi Row, and Correlated Subqueries with EXISTS/ANY/ALL
`MYSQL_VIEWS_SIMPLE_COMPLEX_AND_TRIGGERS.sql`	Simple & Complex Views, BEFORE/AFTER INSERT/UPDATE/DELETE Triggers

3. Topic Breakdown

3.1 Clauses & Operators — ECommerceDB

Database: ECommerceDB | Tables: Product, Sales

Clauses Covered

- SELECT DISTINCT — retrieve unique values
- AS — column aliases
- WHERE — filter rows

- ORDER BY — sort results
- LIMIT — restrict row count

Operators Covered

- **=, <>, >, <, >=, <=**
- **+, -, *, /**
- **AND, OR, NOT**
- **IN, NOT IN, BETWEEN, NOT BETWEEN, LIKE, NOT LIKE, IS NULL, IS NOT NULL**

Sample Query

```
SELECT product_name, price, price + (price * 0.10) AS Increased_Price
FROM Product
WHERE price BETWEEN 1000 AND 20000 AND product_name LIKE 'M%';
```

3.2 Aggregate Functions, GROUP BY & Window Functions — ECommerceDB

Aggregate Functions

- COUNT(*) — total rows
- SUM() — total of a column
- AVG() — average value
- MAX() / MIN() — highest and lowest values

GROUP BY & HAVING

Groups rows by a column and filters groups using HAVING (equivalent of WHERE for aggregated data).

Window Functions

RANK() OVER (ORDER BY ...) — assigns a rank to each row without collapsing the result set.

```
SELECT product_id, product_name, price,
       RANK() OVER (ORDER BY price DESC) AS product_rank
FROM Product;
```

3.3 DDL Commands — Hospital Database

Database: Hospital | Table: Patients / Patient_Info

Commands Demonstrated

- CREATE DATABASE / CREATE TABLE — with data types and PRIMARY KEY
- ALTER TABLE ADD — add a new column (DoctorAssigned)
- ALTER TABLE MODIFY — change column data type or size
- RENAME TABLE — rename Patients to Patient_Info
- TRUNCATE TABLE — remove all rows but keep structure
- DROP TABLE — completely remove a table

3.4 DML & Constraints — Online Bookstore

Database: Bookstore | Tables: Books, Orders

DML Operations

- INSERT INTO — add new records
- UPDATE SET — modify existing records
- DELETE FROM — remove specific records
- TRUNCATE — clear all data from a table

Constraints Used

- PRIMARY KEY — unique row identifier
- NOT NULL — prevents empty values
- UNIQUE — ensures no duplicate values (e.g., ISBN)
- CHECK — enforces a condition (Price > 0, Quantity > 0)
- FOREIGN KEY — links Orders.BookID to Books.BookID

3.5 Joins & Built-in Functions — School Database

Database: School | Tables: Students, Enrollments, Courses

Join Types

- INNER JOIN — only matching rows from both tables
- LEFT JOIN — all rows from left + matched rows from right
- RIGHT JOIN — all rows from right + matched rows from left

Built-in Functions

- **ROUND(), ABS(), MOD()**
- **CONCAT(), LENGTH(), REPLACE(), SUBSTRING(), UPPER(), LOWER()**
- **NOW(), DATEDIFF(), DATE_ADD()**

3.6 Subqueries — CompanyDB

Database: CompanyDB | Tables: Employees, Departments

Subquery Types

- Single Row Subquery — returns one value (used with =, >, <)
- Multi Row Subquery — returns multiple values (used with IN, ANY, ALL)
- Correlated Subquery — references the outer query; re-runs for each row

Special Keywords

- EXISTS — checks whether a subquery returns any rows
- ANY — true if any comparison in the subquery is satisfied
- ALL — true only if all comparisons in the subquery are satisfied

3.7 Views & Triggers — CompanyDB

Database: CompanyDB | Tables: Employees, Departments + Audit/Log tables

Views

- Simple View — selects from a single table (EmployeeBasicView)
- Complex View — joins multiple tables (EmployeeDepartmentView)
- Aggregated View — uses GROUP BY + JOIN (DeptSalaryStats)
- Updatable View — UPDATE through a view reflects to the base table
- DROP VIEW — removes a view definition

Triggers

- BEFORE INSERT — validate salary >= 30,000 before inserting
- AFTER INSERT — log new employee to EmployeeAudit table
- AFTER UPDATE — log salary changes to SalaryLog table
- BEFORE DELETE — block deletion of IT department employees
- AFTER DELETE — archive deleted employees to EmployeeArchive table

4. Databases Summary

Database	Tables	Key Topics
ECommerceDB	`Product, Sales`	Clauses, Operators, Aggregates, Window Functions
Hospital	`Patient_Info`	DDL (CREATE, ALTER, RENAME, TRUNCATE, DROP)
Bookstore	`Books, Orders`	DML, Constraints (PK, FK, UNIQUE, CHECK, NOT NULL)
School	`Students, Enrollments, Courses`	Joins, Built-in Functions
CompanyDB	`Employees, Departments + Logs`	Subqueries, Views, Triggers

5. How to Run the Scripts

Prerequisites

- MySQL Server 8.0 or higher installed
- MySQL Workbench, DBeaver, or any MySQL-compatible client

- Basic familiarity with running SQL scripts

Steps

- Open your MySQL client and connect to your server.
- Open any .sql file from this project.
- Run the CREATE DATABASE and USE statements first to set up the target database.
- Execute the remaining statements in order from top to bottom.
- For Subqueries and Views & Triggers files, ensure the CompanyDB tables (Employees, Departments) are populated with data first.

Recommended Study Order

Follow this sequence for a logical learning progression:

Step	File
1	DDL Commands (Hospital)
2	DML & Constraints (Bookstore)
3	Clauses & Operators (ECommerceDB)
4	ORDER BY, Aggregates & Window Functions
5	Joins & Built-in Functions (School)
6	Subqueries (CompanyDB)
7	Views & Triggers (CompanyDB)

6. Key SQL Concepts Quick Reference

DDL vs DML

- **CREATE, ALTER, DROP, TRUNCATE, RENAME — define structure**
- **INSERT, UPDATE, DELETE, SELECT — manipulate data**

Joins vs Subqueries

- **combine rows from two or more tables horizontally**
- **nest a SELECT inside another query for filtered or computed values**

Views vs Triggers

- **virtual tables based on a SELECT query; simplify complex queries**
- **automatic actions that fire on INSERT, UPDATE, or DELETE events**

7. Notes & Best Practices

- Always run USE <database_name>; before executing any table-level queries.
- In Subqueries and Views/Triggers files, manually insert sample data into Employees and Departments before running SELECT queries.
- Triggers in Section 7 assume the Employees and Departments tables already exist.
- The BEFORE DELETE trigger blocks deletion of IT department employees — test with non-IT employees.
- Window functions (RANK, ROW_NUMBER, etc.) require MySQL 8.0 or later.
- CHECK constraints are fully enforced from MySQL 8.0.16 onwards.